Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/20/2013
PI Name:	Karma, Alain Ph.D.		
Project Title:	Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys		
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Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
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PI Email:	a.karma@neu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617 373 2929
Organization Name:	Northeastern University		
PI Address 1:	Department of Physics		
PI Address 2:	360 Huntington Ave		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02115-5005	**Congressional District:**	8
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Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
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	Cellular and dendritic microstructures formed during industrial solidification processes (such as casting, welding, and brazing) determine the properties of commercial metallic alloys produced in numerous industries, with applications ranging from automobile engine parts to aircraft turbine blades. From a fundamental standpoint, the formation of these structures is controlled by a subtle interplay between microscopic interfacial phenomena and macroscopic transport of heat and solute, which is strongly influenced by gravity-induced melt convection under terrestrial conditions. Therefore, microgravity directional Bridgman experiments provide a unique means to investigate microstructure development with negligible convection and to test the predictions of theoretical models in a purely diffusive growth regime. While most of the theoretical research to date has focused on the growth of single crystals, most commercially relevant metallic alloys are polycrystalline. The main aim of this research is to bring a quantum leap in our ability to model		

Task Description:	realistically dendritic alloy structures in polycrystalline materials where interactions between well-developed dendrites from the same grain and from grains of different crystal orientations both need to be considered. Interactions at grain boundaries are especially important because of their influence on dendrite fragmentation and the formation of solidification defects. This aim will be pursued by developing a new coarse grained dendritic network approach that is capable of simulating quantitatively the long-range diffusive interactions between primary, secondary and higher order branches of typical hierarchical dendritic tree structures on experimentally relevant length and time scales. Validation of this approach will be carried out through quantitative comparisons with phase-field simulations as well as both microgravity and ground based experiments in well-characterized aluminum based binary alloys, such as Al-Cu and Al-Si. Those studies will set the stage for planning future microgravity Bridgman experiments on the International Space Station that use bi-crystalline samples to investigate both intragrain and intergrain dendrite interactions in a diffusive growth regime.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to light weight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. This research is aimed at developing computational tools to predict reliably the formation of these dendritic microstructures on experimentally relevant length and time scales so as to accelerate and optimize the design of new technological alloys that can directly benefit US industries.
Task Progress:	We have developed a novel dendritic needle network (DNN) model capable of simulating quantitatively the solidification of concentrated dendritic alloys with well developed primary, secondary, and tertiary branches. This model makes it possible to simulate accurately the processing conditions used to solidify Al based alloy samples in microgravity experiments. This is achieved by reliably bridging the gap between phase-field simulations on the scale of the dendrite tip radius and cellular-automaton simulations at the much larger scale of several dendritic grains. The DNN model represents a dendritic network of primary, secondary, and higher other branches with a network of sharp needles interacting through the solutal diffusion field. The growth velocity V of each needle tip is computed by combining a standard solvability condition that fixes the product rho2V and an additional solutal flux balance condition that states that the product rho2V is proportional to F^2 where F is a measure of the solute flux received by the tip region. In 2D, this solutal flux intensity factor (FIF) can be accurately computed by a contour integral similar to the J-integral commonly used in fracture mechanics to compute stress intensity factors at the tip of a crack. The DNN formulation provides an asymptotically exact description of the dendritic network dynamics in the limit of small Péclet number rhoV/D << 1, where D is the solute diffusivity. We developed the DNN model for both isothermal and directional solidification in two dimensions. We validated the prediction of the model with comparisons with analytical predictions for early-stage and steady-state dendritic growth kinetics, and with comparison with quantitative phase-field simulations. The comparison with the PF model revealed that the DNN simulations are four orders of magnitude faster than PF simulations in 2D, while keeping a reasonable accuracy as long as the tip Péclet number remains small (e.g. a relative error of ~25% on the early-stage dendrite tip velocity for a dimensionless chemical supersaturation of 0.1, i.e. a Péclet number ~0.003). The DNN approach was extended to non-isothermal growth under directional solidification conditions. We used the DNN model to simulate the growth of a spatially extended dendritic array. For the Al-7wt%Si alloy processed in the CETSOL (Columnar-to-Equiaxed Transition in Solidification processing – European Space Agency project) microgravity experiments, we studied the stability and history-dependent dynamical selection of dendritic spacing. For given solidification conditions close to the experiment, we calculated the stable range of primary dendritic spacing. We also observed that the selected primary spacing after a planar interface destabilization tends to be lower than after an abrupt change of the sample cross section. This is to our knowledge the first time that such important microstructural features resulting from the diffusive interaction of primary, secondary and higher order branches can be directly simulated on a centimeter scale for realistic alloy and processing parameters. We also compared DNN predictions to microgravity experiments for the same alloy, processing condition and sample size, hence showing a reasonable agreement for the selected primary dendritic spacing. Additionally, we developed a massively parallel implementation on Graphic Processing Units (GPUs) of a quantitative phase-field model for directional solidification previously presented and validated in the literature. The parallelized simulations were found to run 150 times faster than standard single core phase-field calculations. Then, we used the computationally efficient PF implementation to perform bi-crystalline phase-field simulations at length and time scales relevant for experiments. We achieved several for a succinonitrile (SCN)-acetone alloy, which has been thoroughly investigated experimentally. This has enabled us to run numerous computational experiments and investigate the influence of several control parameters, including the grain misorientation alpha 0, the growth rate V, and the temperature gradient G, on the selection of dendritic spacing and on the macroscopic evolution of grain boundaries. In terms of the selected dendritic microstructure within each grain, our simulations confirmed the proposed scaling laws for influence of G and a on primary dendritic spacing, with a remarkable quantitative agreement using parameters identified in the literature. We also confirmed the existence of a master curve for the dependence of the dendrite growth direction versus the crystal orientation as an only function of the spacing Péclet number. We observed that the effect of the temperature gradient on the dendritic spacing appears more important than the effect of the crystal orientation. Also, the selection of spacing in the well-oriented grain does not seem to depend on the orientation of the neighboring grain, and thus follows the same scaling law. In terms of grain competition, we observed that the microscopic fluctuations, which are amplified on a macroscopic scale to produce side branches, have a crucial influence on the final direction of the macroscopic grain boundary (GB) between misoriented grains. Hence, a similar simulation with identical control parameters but different fluctuations may give rise to significantly different survival rate of a misoriented grain. Even in two dimensions, tertiary branching at a diverging GB is manifested as a stochastic event that is not periodic in time. As opposed to the classical grain selection theory by Walton and Chalmers, a higher crystal misorientation, which leads to a higher undercooling, does not systematically lead to a faster elimination of the misoriented grain. Indeed, the evolution of the macroscopic grain boundary angle with the crystal misorientation angle is not monotonous and a wide range of misorientation may lead to a similar average

	elimination rate. Our simulations also confirmed the existence of an unexpected overgrowth mechanism of the well-oriented grain by the misoriented grain, which was only observed in some simulations for low grain misorientation and low temperature gradient. We also observed that, when the misoriented grain grows with a seaweed structure (i.e. a structure without preferred primary branch growth directions) at high grain misorientation and high temperature gradient, the favorably oriented grain overgrows the seaweed structure much faster than a dendritic structure.
Bibliography Type:	Description: (Last Updated: 05/10/2026)
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